In-class Activity

EK 505 - Prof. Tron

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For the question tagged with the label **image** require you need to submit an image on Gradescope for evaluation.

Problem 1: Reference frames with `spatialmath`

Preparation. Start a terminal in Visual Studio Code (make sure to have the right environment selected), and start `ipython`. Import the class `SE3` from the module `spatialmath`.

Question image 1.1. Create a transformation `T01` that is a rotation around the z axis (`SE3.Rz()`) followed by a translation along the z axis (`SE3.Tz()`). Create a transformation `T12` that is a rotation around the x axis (`SE3.Rx()`) followed by a translation along the z axis (`SE3.Tz()`). Visualize `T01`, `T12`, and their product `T01*T12`

Problem 2: Swift sim

Preparation. Start a new Python file. Add the following lines to import the file `spherical.urdf` from the current directory:

```
from os import path
from roboticstoolbox.models.URDF.URDFRobot import URDFRobot

robot = URDFRobot(path.join(path.abspath(path.curdir), 'spherical.urdf
    '))
```

Question image 2.1. Add an import for the `swift` module. Use the following loop to visualize the robot. Change the joint angles to see the effect.

```
env = swift.Swift()
env.launch(realtime=True)
env.add(robot)

while True:
    robot.q = [0,0]
    env.step(0.05)
```

Question optional 2.1. Create another loop so the robot joint angles are time-varying.